Report 1: Initial Kagome-VQE Project Note

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Potential solution. This project explores a no-calibration route for preparing low-energy states of the 19-site Kagome antiferromagnetic Heisenberg benchmark. Instead of changing the Hamiltonian, the idea is to start from a physically motivated signed weighted-RVB state and then refine it with shallow edge-colored Heisenberg-HVA circuit layers. The exact benchmark is

$$E_0 = -29.146168109350,$$

while the current preliminary no-calibration result reaches

$$E = -29.037601351802, \qquad F = 0.982518440088$$

at HVA depth $p = 4$. This is not a final proof of scalable quantum spin liquid preparation; it is an initial finite-size proof-of-concept showing that better state structure plus Heisenberg-level circuit refinement can move a dimer/RVB state close to the exact 19-site ground state.

Sketch.

$\boxed{\text{19-site Kagome bonds}} \rightarrow \boxed{\text{signed weighted RVB}} \rightarrow \boxed{\text{Heisenberg-HVA}} \rightarrow \boxed{\text{energy/fidelity check}}$

Current interpretation. The weighted-RVB initializer captures most of the useful finite-size state structure, and the Heisenberg-HVA layers give a smaller but real circuit-compatible refinement. The calibrated-Hamiltonian work discussed in the project provides useful methodology context and clues for future ansatz design, but the main current direction is to keep the target Hamiltonian unchanged and improve the circuit/state ansatz.

Repository / current work link. https://github.com/adoolelomani2026/qintern-kagome-hva

Next step. The next controlled experiment is to test calibration-informed local circuit blocks, such as adding extra Heisenberg gates on the important triangle $(6, 7, 8)$ or bond $(14, 16)$, while still evaluating against the original unmodified Hamiltonian.